

Outdoor Enclosure for Camera Wiring

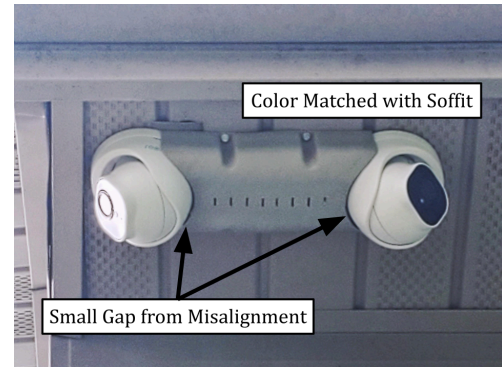
Designer and Fabricator / Apr 2025

Overview:

Designed and fabricated a weather-resistant, 3D-printed enclosure to conceal and protect security camera wiring. The enclosure was color matched to blend with the environment, included drainage features, and was precisely dimensioned for easy retrofit onto existing installations. Installation insights highlighted the importance of camera placement and preparation for efficient assembly.

Tools & Skills:

Fusion 360 (3D modeling), PrusaSlicer (3D printing preparation), Design for Additive Manufacturing, Practical Installation Planning



Installed enclosure at camera site.

Objective:

Design and fabricate a weather-resistant enclosure to conceal and protect the wiring of installed security cameras.

Design Highlights:

- Color matched to the surrounding environment to minimize visual impact and reduce attention.
- Optimized for 3D printing, ensuring efficient material use and ease of fabrication.
- Integrated weep holes to enable rapid drainage, preventing water accumulation in the unlikely event of ingress.
- Precisely dimensioned to fit existing camera installations with minimal tolerances, allowing for straightforward retrofit without modification.

Reflections:

The sheltered mounting location under a soffit reduced the need for heavy-duty weatherproofing and impact resistance, enabling a streamlined enclosure design optimized for 3D printing and ease of installation. The 3D-printed part's surface finish was below expectations due to a loose printer component, though it remained functional. Installation was generally straightforward, but the elevated mounting location required a large ladder, adding logistical difficulty. A printed template for hole locations could have allowed for pre-drilling, simplifying the process and reducing time at height. A key lesson learned was the importance of installation order: installing the cameras before the enclosure introduced alignment challenges, resulting in a slight misfit and visible gap that could have been avoided by reversing the sequence.